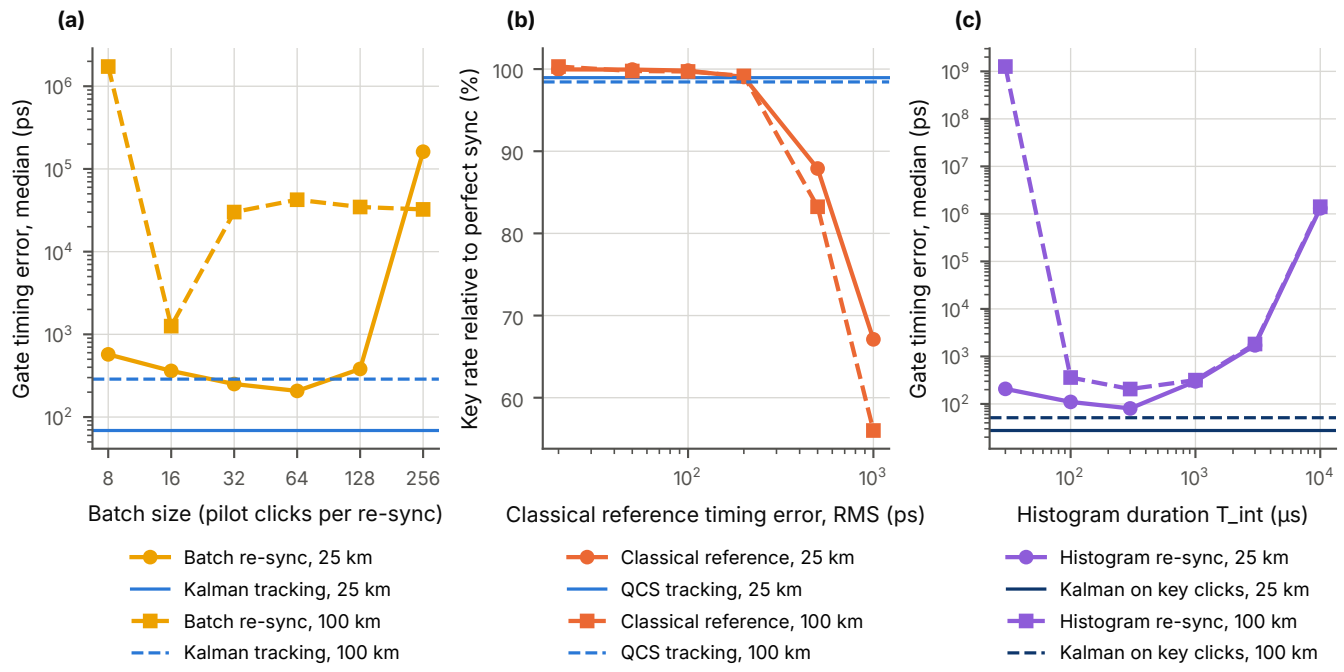
(a)
Gate timing error, median (ps)
10^6
10^5
10^4
10^3
10^2
8
16
32
64
128
256
Batch size (pilot clicks per re-sync)
Batch re-sync, 25 km
Kalman tracking, 25 km
Batch re-sync, 100 km
Kalman tracking, 100 km
(b)
Key rate relative to perfect sync (%)
100
90
80
70
60
10^2
10^3
Classical reference timing error, RMS (ps)
Classical reference, 25 km
QCS tracking, 25 km
Classical reference, 100 km
QCS tracking, 100 km
(c)
Gate timing error, median (ps)
10^9
10^8
10^7
10^6
10^5
10^4
10^3
10^2
10^2
10^3
10^4
Histogram duration T_int (µs)
Histogram re-sync, 25 km
Kalman on key clicks, 25 km
Histogram re-sync, 100 km
Kalman on key clicks, 100 km